0.5 Project Scope Sheet · Group 1 · Fernão de Oliveira

11 November 2026 · T. Santos, L. Batista

Ship and Ocean Systems Design 2026 · Ship Design Project · Scope sheet for your group

Field	Value
Group	Group 1 · Fernão de Oliveira
Members (student number)	106238 · 107304 · 106645
Baseline	B · Azores Feeder: geared container ship, 130 TEU, battery-electric with swappable battery containers, two island loops plus a transit each week
Specialization 1	SP01 · Longitudinal Strength Calculation
Specialization 2	SP02 · Propulsion Matching
Deadline	Dec 13, 2026, one ZIP file by e-mail to luis@veranavis.com
ZIP name	Group1_Project.zip
Presentation	Dec 18, 2026, 15 minutes per group

Deliverable Groups D1–D11

#	Deliverable	Your project must deliver (group, outside class)
D1	Design Basis	130 TEU at 10 t per TEU (1,300 t cargo) · battery containers as extra deadweight, mass from supplier data · 11–12 kn · two loops plus the transit in a week · limits at all eight ports, each with its source · every requirement tagged Given / Assumed / To be checked
D2	Reference Ship Database	At least 8 small geared container ships and battery ships · other families as your chosen hull needs · sources for every ship · your own regressions and bounds
D3	Ship Model	Volume-carrier sequence · inputs, outputs and calculation sequence · open-water propeller efficiency η0 as a stated, justified assumption (0.60) · electric drive, two shafts · batteries as deadweight · crane in the lightship · closed weight balance
D4	Voyage and Financial Model	Both loops and the transit: energy per leg, battery swaps and charging, electricity and port costs · building cost, financing, RFR and NPV · ship model outputs as input
D5	Validation Note	Your model against the reference families you chose, with the choice justified · error per quantity · model corrected where needed
D6	Optimization Report	Minimum battery capacity that completes both loops in one week · justified objective · active constraints · optimum against the requirements
D7	Compliance Sheet	EEDI and CII: show why they do not apply, and report energy per TEU-mile · battery rules (BV) · stability while the crane lifts · freeboard · draught at every port
D8	General Arrangement and Hull Form	Deck container layout and crane reach · battery slots aft · hull form of your choice, justified (a PSV-type hull is only a suggestion) · A3 sketch (profile and midship section) · 3D hull shape with the containers in the holds and on deck, the main cargo-hold bulkheads, the collision bulkhead and the deckhouse exterior · hydrostatics check
D9	Specialization Documents (2)	One individual report per specialization, with its required studies and drawings (see Your specializations)
D10	Datasheet and Presentation	One-page A4 datasheet (0.4 Datasheet Template) · report on the 0.3 Report Template · presentation of 15 minutes
D11	Code Flowchart (mandatory)	A flowchart of your code that shows how the information flows: inputs, each calculation step, the iterations and the outputs · symbols of ISO 5807, the standard for data, program and system flowcharts

Your studies and drawings

Your group's list from the Statement of Work: 34 items, 6 of them drawings. Tick each one off as it is done.

ID	Title	Deliverable	Status
STU-D1.1	Owner's requirements and port limits	D1	
STU-D1.2	Environmental targets	D1	
STU-D2.1	Reference ship database with sources	D2	
STU-D2.2	Regressions and bounds	D2	
STU-D3.1	Ship model	D3	
STU-D3.2	Closed weight balance	D3	
STU-D4.1	Voyage model	D4	
STU-D4.2	Financial model	D4	
STU-D5.1	Validation against reference ships	D5	
STU-D6.1	Optimization	D6	
STU-D7.1	EEDI and CII	D7	
STU-D7.2	Freeboard and intact stability	D7	
STU-D7.3	Draught at every port	D7	
STU-D8.1	Hydrostatics check	D8	
DWG-D8.1	Profile (A3)	D8	
DWG-D8.2	Midship section (A3)	D8	
DWG-D8.3	3D hull with containers, bulkheads and deckhouse	D8	
STU-D10.1	Datasheet (template 0.4)	D10	
STU-D10.2	Presentation, 15 minutes	D10	
STU-D11.1	Code flowchart (ISO 5807)	D11	
STU-B.1	Capacity count on own arrangement	D8	
DWG-B.1	Container and battery layout with crane reach	D8	
STU-SP01.1	Lightship and deadweight distribution	D9 · SP01	
STU-SP01.2	Four critical loading conditions	D9 · SP01	
STU-SP01.3	Still-water and wave bending moments and shear forces	D9 · SP01	
STU-SP01.4	Allowable stresses, bending and shear	D9 · SP01	
STU-SP01.5	Preliminary scantlings of the midship cross-section	D9 · SP01	
STU-SP01.6	Section modulus, moment of inertia and neutral axis	D9 · SP01	
STU-SP01.7	Allowable still-water bending moment and shear force vs loading conditions	D9 · SP01	
DWG-SP01.1	Midship cross-section	D9 · SP01	
STU-SP02.1	Optimum B-series propeller	D9 · SP02	
STU-SP02.2	Engine matching	D9 · SP02	
STU-SP02.3	PTO/PTI or electric operating modes	D9 · SP02	
DWG-SP02.1	Stern arrangement	D9 · SP02	

Your specializations

One calculation note per specialization goes in D9, with the drawings listed. Each specialization has its own self-study deck.

SP01 · Longitudinal Strength Calculation

Show that the hull girder carries the still-water and wave bending moments.

- STU-SP01.1 Lightship weight and deadweight distribution along the length.
- STU-SP01.2 Four critical loading conditions (subject to justification and/or definition with the instructor).
- STU-SP01.3 Still-water and wave bending moments and shear forces for each loading condition.
- STU-SP01.4 Allowable stresses, in bending and in shear.
- STU-SP01.5 Preliminary scantlings of the midship cross-section.
- STU-SP01.6 Section modulus, moment of inertia and neutral axis of the midship section.
- STU-SP01.7 Allowable vertical still-water bending moment and shear force: the allowable envelopes along the length, with each loading condition plotted against them.
- DWG-SP01.1 Midship cross-section.

SP02 · Propulsion Matching

Choose the propeller and match it to the motors in every operating mode.

- STU-SP02.1 Optimum Wageningen B-series propeller for the service speed.
- STU-SP02.2 Engine matching: operating points on the motor layout diagram, with the sea margin.
- STU-SP02.3 The two-shaft electric modes (Baseline B).
- DWG-SP02.1 Stern arrangement: propeller, shaft line and rudder.

The full text of every item is in the Statement of Work.